Rayen Ben Fathallah — AI engineer

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github.com/rayenFathallah Tunis, Tunisia

Summary

AI Engineer with 2+ years of experience building and delivering data driven & AI solutions in production. Experienced in taking projects from development to deployment, with a focus on practical, reliable solutions and real-world impact.

Education

Faculty of Sciences of Tunis Sept 2020 – Jun 2025
National Engineering Diploma in Applied Sciences and Technologies Specialization: Data Science *Tunis, Tunisia*

Experience

Rosenberger July 2026 - present
Senior Data & AI engineer *Sousse, Tunisia*

- Designed and deployed AI solutions for industrial manufacturing, leveraging computer vision, machine learning, and optimization to improve quality inspection and production processes.
- Led AI engineering initiatives from problem definition to production, developing data pipelines, ML models, and decision-support tools while collaborating with business and engineering teams.

Carbon Insights June 2025 – June 2026
Data Scientist *Tunis, Tunisia*

- Built and delivered 4 AI products for European energy and carbon markets, taking solutions from R&D to production for international energy companies.
- Developed a multi-source AI system combining real-time news with market and external data, using context management engines to assess news relevance and horizon-level impact on energy and carbon markets.
- Developed and deployed market analytics models, owning the lifecycle from data sourcing and feature engineering to monitoring, validation, and retraining.

NextShine May 2025 – May 2026
Part-time AI Engineer *Tunis, Tunisia*

- Architected and developed a multi-agent AI platform supporting founders through structured agent workflows from idea validation to startup execution.
- Engineered chat-based and batch LLM pipelines using agent orchestration, context engineering, and multiple inference strategies, evaluating proprietary and local LLMs to optimize cost, latency, and output quality.
- Implemented RAG pipelines for information extraction and context-aware retrieval, and integrated VLLMs to power interactive chatbot experiences with real-time, conversational access to platform knowledge.
- Designed production AI workflows with context management, caching, model routing, evaluation, and observability across use cases to improve inference efficiency and reliability.

Carbon Insights Jan 2025 – June 2025
Data Scientist Intern *Tunis, Tunisia*

- Developed multi-type, multi-horizon time-series forecasting models for power production and electricity demand across European regions, with forecasts up to 2 weeks ahead.

GoMyCode May 2024 – Nov 2024
Data Science Instructor *Online*

- Taught data science topics including machine learning, TensorFlow, and PyTorch.

Projects

Analyzing and Managing Open Data in Peru - Open Source | *Python, Pandas, Selenium, Data Pipelines, Power BI*

- Built Python data-collection and processing pipelines using Pandas and Selenium, with data-quality validation on public safety datasets, and created interactive Power BI dashboards to visualize crime trends and support policy analysis.

Skills

Programming: Python, SQL, JavaScript, Linux
AI Engineering: LLMs, VLMs, RAG, Agentic AI, AI Agents, LLM Orchestration, Context Engineering, Prompt Engineering, Fine tuning, Model Routing, Inference Optimization, LangChain, LangGraph, Transformers, Hugging Face, NLP, NER, Embeddings, Vector Search, Semantic Search
Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Computer Vision, Time Series Forecasting, Feature Engineering, MLflow
MLOps & Deployment: AWS, Docker, CI/CD, GitHub Actions, Model Deployment, Model Monitoring, Drift Detection, Model Versioning
Data & Backend: PostgreSQL, Pandas, NumPy, ETL, Data Pipelines, Data Quality, FastAPI, Flask, REST APIs, Redis